

Introduction

- Accurate height measurement is essential for monitoring childhood **stunting**.
- Demographic and Health Surveys (DHS)** are a major global source of child growth data, but often contain missing or implausible height measurements.
- Missing height data can create **bias in estimating stunting prevalence**, especially when **systematically missing rather than random**.
- Understanding these patterns is crucial before applying imputation or predictive modeling approaches.
- This study examines **missingness patterns** in DHS anthropometric data across **Bangladesh, India, Nepal, and Pakistan (2011–2022)**.

Objectives

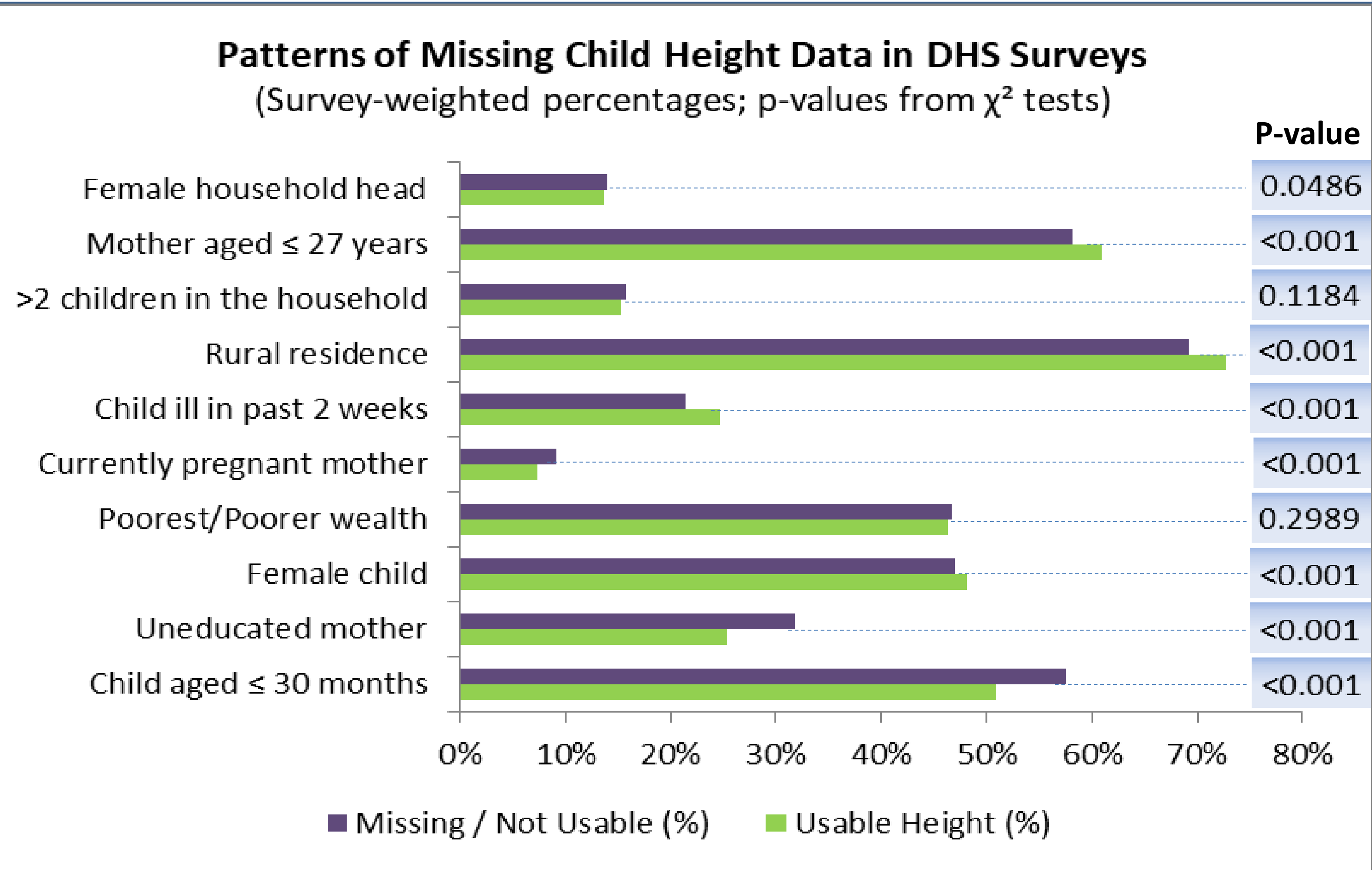
- Assess the **extent and patterns of missing child height** measurements using **height-for-age-Z-scores (HAZ)**.
- Identify **key predictors of missingness** using survey-weighted methods.
- Explore whether **imputation** can help recover missing HAZ values.

Data and Methods

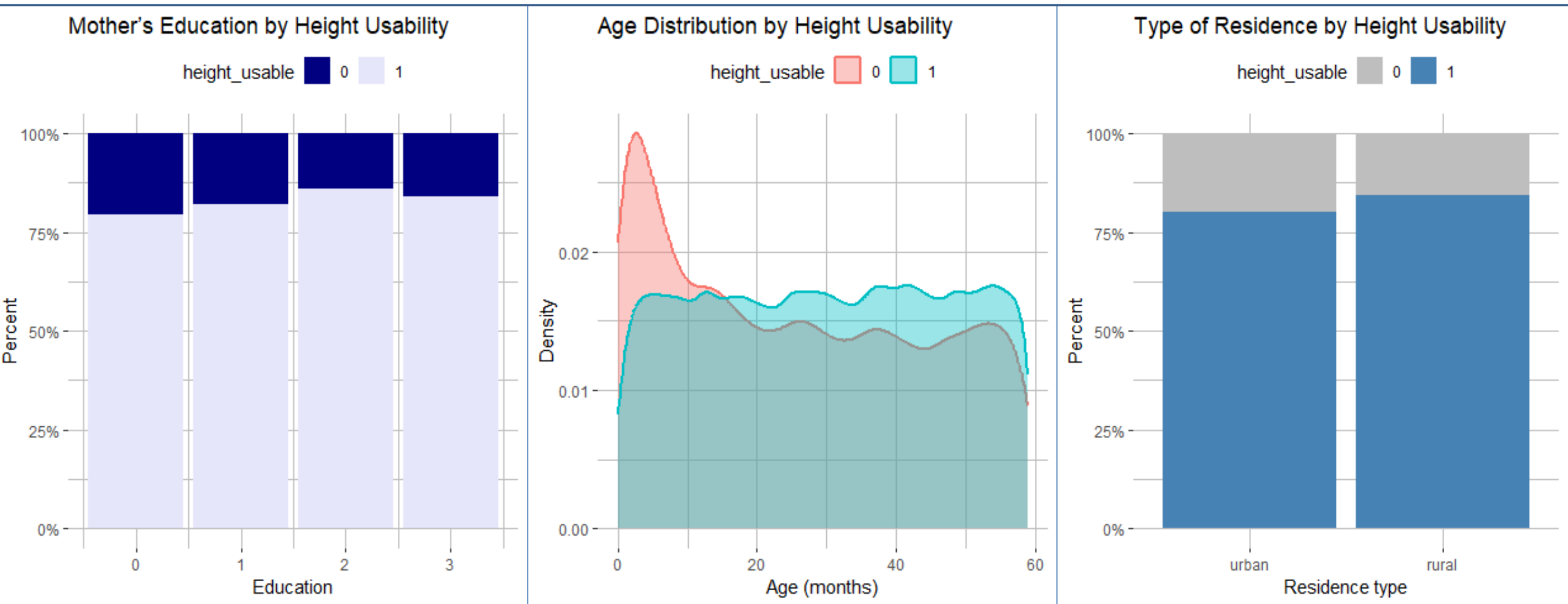
- ☐ **Datasets**
- Pooled **11 DHS surveys (2011–2022)** from **Bangladesh, India, Nepal, and Pakistan** (total sample: **≈553,000 children under 5**).
- ☐ **Outcome Variables**
- Height usability** (usable = 1 versus missing or not plausible = 0), **HAZ** (WHO reference).
- ☐ **Analytical Approach**
- Survey-weighted χ^2 tests and t-tests** to compare children with and without usable data.
 - Survey-weighted multivariable logistic regression** to identify predictors of missingness.
 - Missingness pattern assessment** to evaluate the plausibility of the **Missing at Random (MAR)** assumption.
 - Predictive Mean Matching (PMM)** to impute missing anthropometric outcomes.
- ☐ **Imputation diagnostics**
- Density plots comparing observed and imputed HAZ.
 - Summary statistics across multiple imputed datasets.

Results and Interpretations

- ☐ **Extent of Missing Height Data:**
- Across 553,000 children from 11 DHS surveys, **approximately 17% of children had missing or non-plausible height measurements**.
 - The prevalence of missing height data was **similar but slightly lower in 2017–2022** compared with 2011–2016.
- ☐ **Key Descriptive Differences (Usable and Missing or Implausible Height Data):**
- Missing height measurements are **systematically patterned**, not randomly distributed across the population.



- Children ≤ 30 months** were significantly more likely to have **missing or non-usable height measurements**.
 - Missing height data were more common among children of **uneducated mothers**.
 - Rural residence, recent child illness, and maternal pregnancy** were also associated with higher missingness.
- The bivariate plots of height usability with observed predictors also show evidence of **socio-demographic differences by height measurement availability**.



- ☐ **Predictors of Missing or Implausible Child Height Measurements:**
- Survey-weighted multivariable logistic regression:** outcome(height usability) and predictors (socio-economic and demographic characteristics).

Characteristic	Overall Adjusted Odds Ratio (OR) (95% CI)	2011–2016 (OR)	2017–2022 (OR)
Overall negatively associated			
Child age ≤ 30 months	0.739*** (0.716–0.764)	0.712***	0.772***
Mother with no education	0.892*** (0.855–0.930)	0.896***	0.912**
Overall positively associated			
Female child	1.078*** (1.044–1.113)	1.115***	1.035
Poorest/Poorer wealth	1.109*** (1.068–1.152)	1.110***	1.102**
Mother pregnant	1.227*** (1.147–1.311)	1.298***	1.146**
Child ill in past 2 weeks	1.336*** (1.284–1.391)	1.332***	1.345***
Rural residence	1.427*** (1.358–1.499)	1.363***	1.498***
Overall not associated			
>2 children in household	0.953 (0.907–1.001)	0.987	0.913*
Mother age ≤ 27 years	1.018 (0.978–1.059)	0.996	1.051
Female household head	1.039 (0.988–1.092)	1.071*	0.997

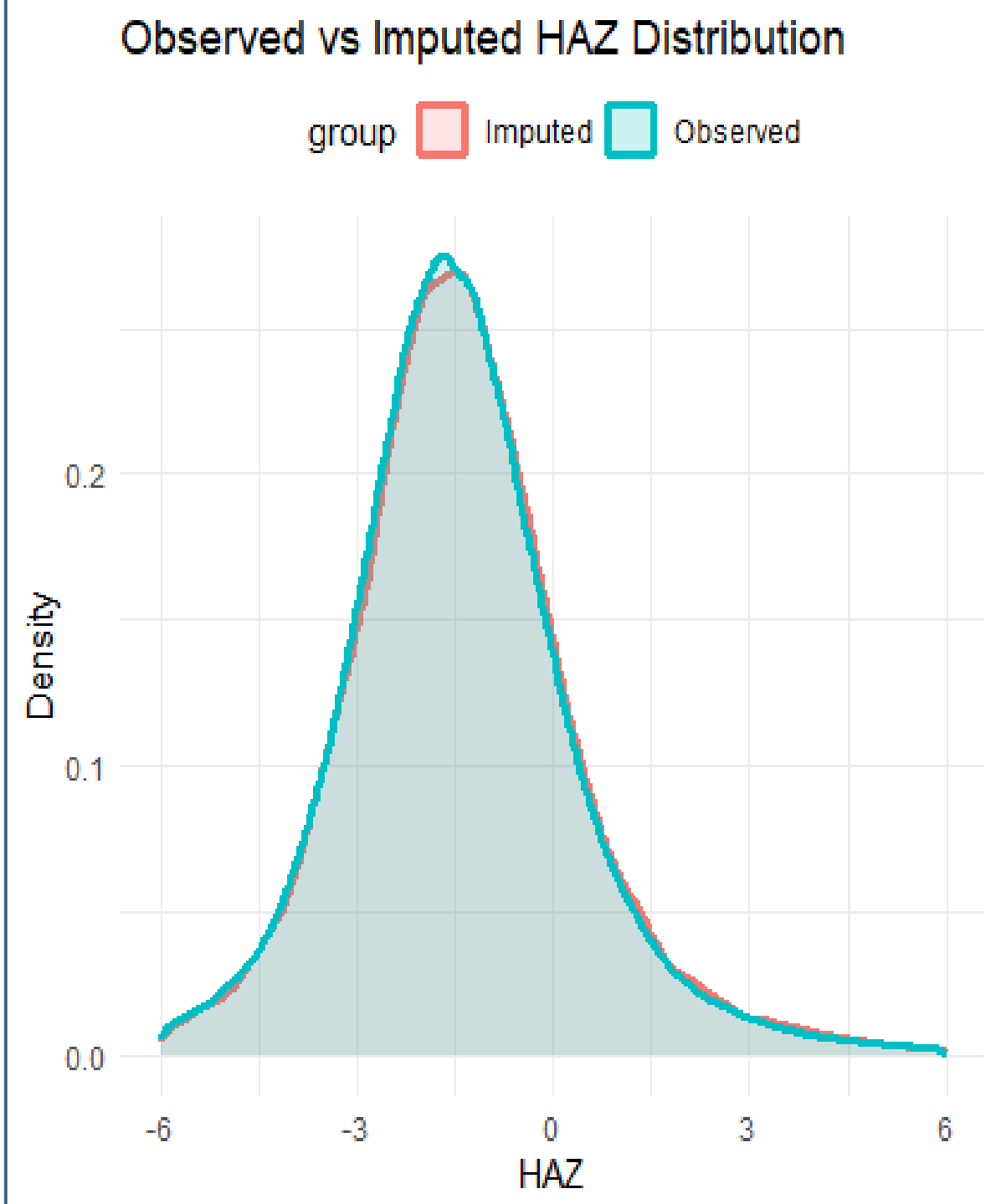
Note: Significance levels are shown as * (p < 0.05), ** (p < 0.01), and *** (p < 0.001).

Observations	509,006	268,954	240,052
Clusters	5,416	2,658	2,758
Percentage affected	16.72%	17.08%	16.32%

- Younger children (≤30 months)** were **~26% more likely** to have **missing** height data.
- Rural residence** was associated with **~43% higher odds** of having **usable** height measurements than urban residence.
- Other strong predictors:** maternal education, current pregnancy, child sex, recent illness, and household wealth.
- Most associations remained **stable across survey periods**, indicating persistent measurement challenges, though some **attenuated in recent years**.
- Missingness pattern:** non-random and predictable, supporting **MAR assumption**.
- Implication:** multiple imputation may effectively recover missing HAZ values, reducing bias in stunting estimates.

Handling Missing HAZ

- ☐ **Imputation Approach:**
- Multiple imputation (MICE)** with **Predictive Mean Matching (PMM)**.
 - Predictors:** child, maternal, and household characteristics.
 - Survey design accounted for in the analysis stage.
- ☐ **Density plot: Observed and Imputed HAZ**
- Imputed HAZ distributions **closely overlapped** with observed values, supporting imputation validity.
 - Mean (≈ -1.40) and variability ($SD = 1.73$) of HAZ remained **stable across imputations**.
- ☐ **Contribution:**
- Reduces bias due to missing anthropometry and retains children otherwise excluded from analysis.



Conclusion

- Missing child height data in South Asian DHS surveys are **systematic rather than random**.
- Missingness is strongly associated with socio-economic and demographic covariates.
- **Systematic missingness can bias estimates of stunting prevalence**, potentially leading to under- or over-estimation.
- Ignoring missing height data may distort **socioeconomic and demographic gradients in child undernutrition**.
- **Predictive Mean Matching (PMM)** helps recover plausible HAZ values while preserving the observed distribution.
- Accounting for missing height data **improves the validity and cross-country comparability** of DHS-based nutrition analyses.

- ☐ **Limitation / scope for work:**
- DHS height data may include measurement and reporting errors.
 - Imputation assumes Missing at Random (MAR) given observed covariates.
 - Country-specific health system differences were not fully explored.
 - Imputed values reflect probable, not observed, measurements.

References

- ☐ Finaret AB, Hutchinson M. Missingness of height data in DHS surveys is not random but has limited impact on stunting estimates. Journal of Nutrition. 2018;148(5):781–789.
- ☐ ICF. Demographic and Health Surveys (DHS), 2011–2022. Rockville, Maryland, USA; National Institute of Population Research and Training (NIPORT) [Bangladesh]; Ministry of Health and Family Welfare; International Institute for Population Sciences (IIPS) [India]; Ministry of Health and Population (MoHP) [Nepal]; National Institute of Population Studies (NIPS) [Pakistan].
- ☐ World Health Organization. WHO Child Growth Standards.
- ☐ Lee KJ, Carlin JB, Simpson JA, Moreno-Betancur M. Multiple imputation for missing data: fully conditional specification versus multivariate normal imputation. International Journal of Epidemiology. 2023;52(4):1268–1275.